

Q1 - 24 June - Shift 1

Let $\lambda x - 2y = \mu$ be a tangent to the hyperbola

$a^2x^2 - y^2 = b^2$. Then $\left(\frac{\lambda}{a}\right)^2 - \left(\frac{\mu}{b}\right)^2$ is equal to:

(A) -2

(B) -4

(C) 2

(D) 4

Space for your notes:

Q2 - 24 June - Shift 2

Let the hyperbola H : $\frac{x^2}{a^2} - y^2 = 1$ and the ellipse

E : $3x^2 + 4y^2 = 12$ be such that the length of latus rectum of H is equal to the length of latus rectum

of E. If e_H and e_E are the eccentricities of H and E

respectively, then the value of $12(e_H^2 + e_E^2)$ is equal to _____.

Space for your notes:

Q3 - 25 June - Shift 2

Let the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

be $\frac{5}{4}$. If the equation of the normal at the point

$\left(\frac{8}{\sqrt{5}}, \frac{12}{5}\right)$ on the hyperbola is $8\sqrt{5}x + \beta y = \lambda$, then

$\lambda - \beta$ is equal to

Space for your notes:

Q4 - 26 June - Shift 2

The normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{9} = 1$ at the point $(8, 3\sqrt{3})$ on it passes through the point :

- (A) $(15, -2\sqrt{3})$ (B) $(9, 2\sqrt{3})$
 (C) $(-1, 9\sqrt{3})$ (D) $(-1, 6\sqrt{3})$

Space for your notes:

Q5 - 26 June - Shift 2

Let a line L_1 be tangent to the hyperbola

$\frac{x^2}{16} - \frac{y^2}{4} = 1$ and let L_2 be the line passing through the origin and perpendicular to L_1 . If the locus of the point of intersection of L_1 and L_2 is $(x^2 + y^2)^2 = \alpha x^2 + \beta y^2$, then $\alpha + \beta$ is equal to _____.

Space for your notes:

Q6 - 28 June - Shift 1

Let the eccentricity of the hyperbola

$H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be $\sqrt{\frac{5}{2}}$ and length of its latus

rectum be $6\sqrt{2}$. If $y = 2x + c$ is a tangent to the hyperbola H, then the value of c^2 is equal to

- (A) 18 (B) 20
 (C) 24 (D) 32

Space for your notes:

Q7 - 28 June - Shift 2

Questions

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Let $a > 0$, $b > 0$. Let e and ℓ respectively be the eccentricity and length of the latus rectum of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Let e' and ℓ' respectively be the eccentricity and length of the latus rectum of its conjugate hyperbola. If $e^2 = \frac{11}{14}\ell$ and $(e')^2 = \frac{11}{8}\ell'$, then the value of $77a + 44b$ is equal to

- (A) 100 (B) 110
(C) 120 (D) 130

*Space for your notes:***Q8 - 29 June - Shift 1**

Let $H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $a > 0$, $b > 0$, be a hyperbola such that the sum of lengths of the transverse and the conjugate axes is $4(2\sqrt{2} + \sqrt{14})$. If the eccentricity H is $\frac{\sqrt{11}}{2}$, then value of $a^2 + b^2$ is equal to _____.

Space for your notes:

Answer Key**Q1 (D)****Q2 (42)****Q3 (85)****Q4 (C)****Q5 (12)****Q6 (B)****Q7 (D)****Q8 (88)**

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Q1 (D) $\lambda x - 2y = \mu$ is a tangent to the curve $a^2x^2 - y^2 = b^2$ then

$$a^2x^2 - \left(\frac{\lambda x - \mu}{2}\right)^2 = b^2$$

$$(4a^2 - \lambda^2)x^2 + 2\lambda\mu x - \mu^2 - 4b^2 = 0$$

Disc. = 0

$$4\lambda^2\mu^2 + 4(4a^2 - \lambda^2)(\mu^2 + 4b^2) = 0$$

$$4\lambda^2b^2 - 4a^2\mu^2 = 16a^2b^2$$

$$\frac{\lambda^2}{a^2} - \frac{\mu^2}{b^2} = 4$$

Q2 (42)

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$$\frac{x^2}{a^2} - \frac{y^2}{1} = 1$$

$$\frac{x^2}{4} + \frac{y^2}{3} = 1$$

$$e_H = \sqrt{1 + \frac{1}{a^2}}$$

$$e_E = \sqrt{1 - \frac{3}{4}} = \frac{1}{2}$$

$$\ell.R. = \frac{2}{a} \quad \ell R = \frac{2 \times 3}{2} = 3$$

$$\frac{2}{a} = 3$$

$$a = \frac{2}{3}$$

$$e_H = \sqrt{1 + \frac{9}{4}} = \frac{\sqrt{13}}{2}$$

$$12(e_H^2 + e_E^2) = 12\left(\frac{13}{4} + \frac{1}{4}\right)$$

$$= \frac{12 \times 14}{4} = 42$$

Q3 (85)

$$e^2 = 1 + \frac{b^2}{a^2} = \frac{25}{16} \Rightarrow \frac{b^2}{a^2} = \frac{9}{16} \dots\dots(1)$$

$$A\left(\frac{8}{\sqrt{5}}, \frac{12}{5}\right) \text{ satisfies } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\Rightarrow \frac{64}{5a^2} - \frac{144}{25b^2} = 1 \dots\dots(2)$$

$$\text{Solving (1) \& (2) } b = \frac{6}{5} \quad a = \frac{8}{5}$$

$$\text{Normal at A is } \frac{\sqrt{5}a^2x}{8} + \frac{5b^2y}{12} = a^2 + b^2$$

$$\text{Comparing it } 8\sqrt{5}x + \beta y = \lambda$$

$$\text{Gives } \lambda = 100, \beta = 15$$

$$\lambda - \beta = 85$$

Q4 (C)

$$\frac{x^2}{a^2} - \frac{y^2}{9} = 1 : (8, 3\sqrt{3}) \text{ lie on Hyperbola then}$$

$$\frac{64}{a^2} - \frac{27}{9} = 1 \Rightarrow a^2 = \frac{64}{4} = 16$$

$$\text{equation of normal at } (8, 3\sqrt{3}) :$$

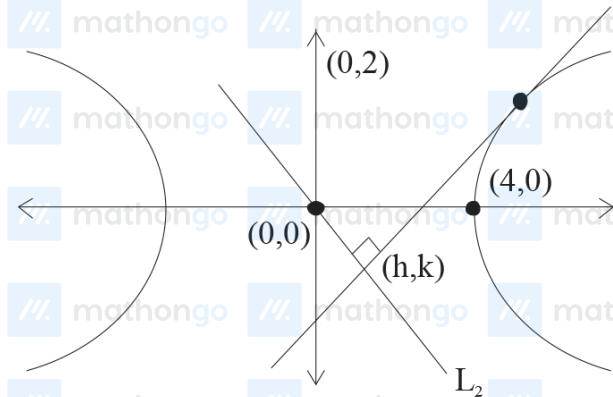
$$\frac{16x}{8} + \frac{9y}{3\sqrt{3}} = 16 + 9$$

$$2x + \sqrt{3}y = 25$$

Check options.

 Q5 (12)					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					

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$$\frac{x \sec \theta}{4} - \frac{y \tan \theta}{2} = 1$$

$$m_1 = \frac{\sec \theta \times 2}{4(\tan \theta)} = \frac{\sec \theta}{2 \tan \theta}$$

$$m_2 = \frac{k}{h}$$

$$m_1 m_2 = -1$$

$$\frac{k}{h} \cdot \frac{\sec \theta}{2 \tan \theta} = -1$$

$$\frac{k}{2h \sin \theta} = -1$$

$$\sin \theta = \frac{-k}{2h} \quad \cos \theta = \frac{\sqrt{4h^2 - k^2}}{2h}$$

also

$$\frac{h \sec \theta}{4} - \frac{k \tan \theta}{2} = 1$$

$$\frac{h}{4} \frac{2h}{\sqrt{4h^2 - k^2}} - \frac{k}{2} \left(\frac{-k}{\sqrt{4h^2 - k^2}} \right) = 1$$

$$h^2 + k^2 = 2\sqrt{4h^2 - k^2}$$

$$(x^2 + y^2)^2 = 4(4x^2 - y^2)$$

$$(x^2 + y^2)^2 = 16x^2 - 4y^2$$

$$\alpha = 16, \beta = -4$$

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$$y = mx \pm \sqrt{a^2 m^2 - b^2}$$

$$m = 2, c^2 = a^2 m^2 - b^2$$

$$c^2 = 4a^2 - b^2$$

$$e^2 = 1 + \frac{b^2}{a^2}$$

$$\frac{5}{2} = 1 + \frac{b^2}{a^2}$$

$$\frac{3}{2} = \frac{b^2}{a^2} \Rightarrow b^2 = \frac{3a^2}{2}$$

$$\frac{2b^2}{a} = 6\sqrt{2}$$

$$\frac{2}{a} \times \frac{3a^2}{2} = 6\sqrt{2}$$

$$3a = 6\sqrt{2}$$

$$\boxed{a = 2\sqrt{2}}$$

$$b^2 = \frac{3}{2} \times 8 = 12$$

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$$b = 2\sqrt{3}$$

$$\therefore c^2 = 4 \times 8 - 12$$

$$c^2 = 20$$

Q7 (D)

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$$e = \sqrt{1 + \frac{b^2}{a^2}}, \ell = \frac{2b^2}{a}$$

$$\text{Given } e^2 = \frac{11}{14} \ell$$

$$1 + \frac{b^2}{a^2} = \frac{11}{14} \cdot \frac{2b^2}{a}$$

$$\frac{a^2 + b^2}{a^2} = \frac{11}{7} \cdot \frac{b^2}{a} \quad \dots\dots(1)$$

$$\text{Also } e' = \sqrt{1 + \frac{a^2}{b^2}}, \ell' = \frac{2a^2}{b}$$

$$\text{Given } (e')^2 = \frac{11}{8} \ell'$$

$$1 + \frac{a^2}{b^2} = \frac{11}{8} \cdot \frac{2a^2}{b}$$

$$\frac{a^2 + b^2}{b^2} = \frac{11}{4} \cdot \frac{a^2}{b} \quad \dots\dots(2)$$

New (1) $\div$ (2)

$$\frac{b^2}{a^2} = \frac{4}{7} \cdot \frac{b^3}{a^3}$$

$$\therefore 7a = 4b \quad \dots\dots(3)$$

From (2)

$$\frac{\frac{16b^2}{49} + b^2}{b^2} = \frac{11}{4} \cdot \frac{16b^2}{49b}$$

$$\frac{65}{49} = \frac{11}{4} \cdot \frac{16}{49} \cdot b$$

$$\therefore b = \frac{4 \times 65}{11 \times 16} \quad \dots\dots(4)$$

We have to find value of

$$77a + 44b$$

$$11(7a + 4b) = 11(4b + 4b) = 11 \times 8b$$

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Hints and Solutions

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Q8 (88)

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\text{Given } e^2 = 1 + \frac{b^2}{a^2} \Rightarrow \frac{11}{4} = 1 + \frac{b^2}{a^2} \Rightarrow b^2 = \frac{7}{4}a^2$$

$$\therefore \frac{x^2}{(a)^2} - \frac{y^2}{\left(\frac{\sqrt{7}}{2}a\right)^2} = 1 \quad \text{Now given}$$

$$2a + 2 \cdot \frac{\sqrt{7}a}{2} = 4(2\sqrt{2} + \sqrt{14})$$

$$a(2 + \sqrt{7}) = 4\sqrt{2}(2 + \sqrt{7})$$

$$a = 4\sqrt{2} \Rightarrow a^2 = 32$$

$$b^2 = \frac{7}{4} \times 16 \times 2 = 56$$

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